TOZIB 5mg Film Coated Tablets

1. NAME OF THE MEDICINAL PRODUCT

TOZIB 5mg Film Coated Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains 8.075mg of tofacitinib citrate equivalent to 5mg tofacitinib.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Film-coated tablet.

White, round, film-coated tablets, debossed with “5” on one side and “T” on the other side.

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

Rheumatoid Arthritis

TOZIB is indicated for the treatment of adult patients with moderately to severely active rheumatoid arthritis (RA) who have had an inadequate response or intolerance to methotrexate. It may be used as monotherapy or in combination with methotrexate or other non-biologic disease-modifying anti-rheumatic drugs (DMARDs).

Limitations of Use: Use of TOZIB in combination with biologic DMARDs or with potent immunosuppressants, such as azathioprine and cyclosporine is not recommended.

Psoriatic Arthritis

TOZIB is indicated for the treatment of adult patients with active psoriatic arthritis (PsA) who have had an inadequate response or intolerance to methotrexate or other disease modifying antirheumatic drugs (DMARDs).

Limitations of Use: Use of TOZIB in combination with biologic DMARDs or with potent immunosuppressants such as azathioprine and cyclosporine is not recommended.

Ulcerative Colitis
TOZIB is indicated for the treatment of adult patients with moderately to severely active ulcerative colitis (UC), who have an inadequate response or who are intolerant to TNF blockers.

Limitations of Use: Use of TOZIB in combination with biological therapies for UC or with potent immunosuppressants such as azathioprine and cyclosporine is not recommended.

4.2 Posology and Method of Administration

Important Administration Instructions

- Do not initiate TOZIB in patients with an absolute lymphocyte count less than 500 cells/mm^3, an absolute neutrophil count (ANC) less than 1000 cells/mm^3 or who have hemoglobin levels less than 9 g/dL.

- Dose interruption is recommended for management of lymphopenia, neutropenia and anemia.
- Interrupt use of TOZIB if a patient develops a serious infection until the infection is controlled.
- Take TOZIB with or without food.

Recommended Dosage in Rheumatoid Arthritis and Psoriatic Arthritis

Table 1 displays the recommended adult daily dosage of TOZIB and dosage adjustments for patients receiving CYP2C19 and/or CYP3A4 inhibitors, in patients with moderate or severe renal impairment (including but not limited to those with severe insufficiency who are undergoing hemodialysis) or moderate hepatic impairment, with lymphopenia, neutropenia, or anemia.

Table 1: Recommended Dosage of TOZIB in Patients with Rheumatoid Arthritis and Psoriatic Arthritis[1]

	TOZIB
Adult patients	5 mg twice daily
Patients receiving: • strong CYP3A4 inhibitors (e.g., ketoconazole), or • a moderate CYP3A4 inhibitor(s) with a strong CYP2C19 inhibitor(s) (e.g., fluconazole)	5 mg once daily
Patients with: • moderate or severe renal impairment • moderate hepatic impairment*	5 mg once daily For patients undergoing hemodialysis, dose should be administered after the dialysis session on dialysis days. If a dose was taken before the dialysis procedure, supplemental doses are not recommended in patients after dialysis.
Patients with lymphocyte count less than 500 cells/mm^3, confirmed by repeat testing	Discontinue dosing.
Patients with ANC 500 to 1000 cells/mm^3	Interrupt dosing. When ANC is greater than 1000, resume 5 mg twice daily.
Patients with ANC less than 500 cells/mm^3	Discontinue dosing.
Patients with hemoglobin less than 8 g/dL or a decrease of more than 2 g/ dL	Interrupt dosing until hemoglobin values have normalized.

[1] TOZIB is used in combination with nonbiologic disease modifying antirheumatic drugs (DMARDs) in psoriatic arthritis. The efficacy of TOZIB as a monotherapy has not been studied in psoriatic arthritis.
* Use of TOZIB in patients with severe hepatic impairment is not recommended.

Recommended Dosage in Ulcerative Colitis

Table 2 displays the recommended adult daily dosage of TOZIB and dosage adjustments for patients receiving CYP2C19 and/or CYP3A4 inhibitors, with moderate or severe renal impairment (including but not limited to those with severe insufficiency who are undergoing hemodialysis) or moderate hepatic impairment, with lymphopenia, neutropenia or anemia.

Table 2: Recommended Dosage of TOZIB in Patients with UC

	TOZIB
Adult patients	**Induction:** 10 mg twice daily for at least 8 weeks; evaluate patients and transition to

	maintenance therapy depending on therapeutic response. If needed continue 10 mg twice daily for a maximum of 16 weeks. Discontinue 10 mg twice daily after 16 weeks if adequate therapeutic response is not achieved. **Maintenance:** 5 mg twice daily. For patients with loss of response during maintenance treatment, a dosage of 10 mg twice daily may be considered and limited to the shortest duration, with careful consideration of the benefits and risks for the individual patient. Use the lowest effective dose needed to maintain response.
Patients receiving: • strong CYP3A4 inhibitors (e.g., ketoconazole), or • a moderate CYP3A4 inhibitor(s) with a strong CYP2C19 inhibitor(s) (e.g., fluconazole)	If taking 10 mg twice daily, reduce to 5 mg twice daily. If taking 5 mg twice daily, reduce to 5 mg once daily.
Patients with: • moderate or severe renal impairment • moderate hepatic impairment*	If taking 10 mg twice daily, reduce to 5 mg twice daily. If taking 5 mg twice daily, reduce to 5 mg once daily.
	For patients undergoing hemodialysis, dose should be administered after the dialysis session on dialysis days. If a dose was taken before the dialysis procedure, supplemental doses are not recommended in patients after dialysis.
Patients with lymphocyte count less than 500 cells/mm^3, confirmed by repeat testing	Discontinue dosing.
Patients with ANC 500 to 1000 cells/mm3	If taking 10 mg twice daily, reduce to 5 mg twice daily. When ANC is greater than 1000, increase to 10 mg twice daily based on clinical response. If taking 5 mg twice daily, interrupt dosing. When ANC is greater than 1000, resume 5 mg twice daily.
Patients with ANC less than 500 cells/mm^3	Discontinue dosing.
Patients with hemoglobin less than 8 g/dL or a decrease of more than 2 g/ dL	Interrupt dosing until hemoglobin values have normalized.

* Use of TOZIB in patients with severe hepatic impairment is not recommended.

Hepatic Impairment

Severe Impairment
Tofacitinib has not been studied in patients with severe hepatic impairment; therefore, use of TOZIB in patients with severe hepatic impairment is not recommended.

Moderate Impairment
Tofacitinib-treated patients with moderate hepatic impairment has greater tofacitinib blood concentration than tofacitinib-treated patients with normal hepatic function. Higher blood concentrations may increase the risk of some adverse reactions. Therefore, dosage adjustment of

TOZIB is recommended in patients with moderate hepatic impairment.

Mild Impairment
No dosage adjustment of TOZIB is required in patients with mild hepatic impairment.

Hepatitis B or C Serology
The safety and efficacy of TOZIB have not been studied in patients with positive hepatitis B virus or hepatitis C virus serology.

Renal Impairment

Moderate and Severe Impairment
Tofacitinib-treated patients with moderate or severe renal impairment has greater tofacitinib blood concentrations than tofacitinib-treated patients with normal renal function. Therefore, dosage adjustment of TOZIB is recommended in patients with moderate or severe renal impairment (including but not limited to those with severe insufficiency who are undergoing hemodialysis).

Mild Impairment
No dosage adjustment is required in patients with mild renal impairment.

Geriatric Use

As there is a higher incidence of infections in the elderly population in general, caution should be used when treating the elderly.

Pediatric Use

The safety and effectiveness of TOZIB in pediatric patients have not been established.

Use in Diabetics

As there is a higher incidence of infection in diabetic population in general, caution should be used when treating patients with diabetes.

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients.

4.4 Special Warnings and Precautions for Use

Serious Infections

Serious and sometimes fatal infections due to bacterial, mycobacterial, invasive fungal, viral, or other opportunistic pathogens have been reported in patients receiving immunomodulatory agents, including biologic DMARDs and tofacitinib. The most common serious infections reported with tofacitinib included pneumonia, cellulitis, herpes zoster, urinary tract infection, diverticulitis, and appendicitis. Among opportunistic infections, tuberculosis and other mycobacterial infections, cryptococcus, histoplasmosis, esophageal candidiasis, multidermatomal herpes zoster, cytomegalovirus infection, BK virus infections, and listeriosis were reported with tofacitinib. Some patients have presented with disseminated rather than localized disease, and rheumatoid arthritis patients were often taking concomitant immunomodulating agents such as methotrexate or corticosteroids which, in addition to rheumatoid arthritis may predispose them to infections. Other serious infections that were not reported in clinical studies, may also occur (e.g., coccidioidomycosis).

In the UC population, TOZIB treatment with 10 mg twice daily was associated with greater risk of serious infections compared to 5 mg twice daily. Additionally, opportunistic herpes zoster infections (including meningoencephalitis, ophthalmologic, and disseminated cutaneous) were seen in patients who were treated with TOZIB 10 mg twice daily.

TOZIB should not be initiated in patients with an active infection, including localized infections. The risks and benefits of treatment should be considered prior to initiating TOZIB in patients with chronic or recurrent infections, or those who have been exposed to tuberculosis, or with a history of a serious or an opportunistic infection or have resided or travelled in areas of endemic tuberculosis or endemic mycoses; or have underlying conditions that may predispose them to infection.

Patients should be closely monitored for the development of signs and symptoms of infection during and after treatment with TOZIB. TOZIB should be interrupted if a patient develops a serious infection, an opportunistic infection, or sepsis. A patient who develops a new infection during treatment with TOZIB should undergo prompt and complete diagnostic testing appropriate for an immunocompromised patient, appropriate antimicrobial therapy should be initiated, and the patient should be closely monitored.

As there is a higher incidence of infections in the elderly and in the diabetic populations in general, caution should be used when treating the elderly and patients with diabetes. Caution is also recommended in patients with a history of chronic lung disease as they may be more prone to infections. Events of interstitial lung disease (some of which had a fatal outcome) have been reported in patients treated with tofacitinib, a Janus-kinase (JAK) inhibitor, in studies and in the post-marketing setting although the role of JAK inhibition in these events is not known.

Risk of infection may be higher with increasing degrees of lymphopenia and consideration should be given to lymphocyte counts when assessing individual patient risk of infection.

Tuberculosis

Patients should be evaluated and tested for latent or active infection prior to and per applicable guidelines during administration of TOZIB.

Patients with latent tuberculosis should be treated with standard antimycobacterial therapy before administering TOZIB.

Antituberculosis therapy should also be considered prior to administration of TOZIB in patients with a history of latent or active tuberculosis in whom an adequate course of treatment cannot be confirmed, and for patients with a negative test for latent tuberculosis but who have risk factors for tuberculosis infection. Consultation with a health care professional with expertise in the treatment of tuberculosis is recommended to aid in the decision about whether initiating antituberculosis therapy is appropriate for an individual patient.

Patients should be closely monitored for the development of signs and symptoms of tuberculosis, including patients who tested negative for latent tuberculosis infection prior to initiating therapy.

Viral Reactivation

Viral reactivation has been reported with DMARD treatment and cases of herpes virus reactivation (e.g., herpes zoster) were observed in studies with tofacitinib. Cases of hepatitis B reactivation have been reported in patients treated with tofacitinib. The impact of tofacitinib on chronic viral hepatitis reactivation is unknown. Screening for viral hepatitis should be performed in accordance with clinical guidelines before starting therapy with TOZIB. The risk of herpes zoster appears to be higher in Japanese and Korean patients treated with tofacitinib.

Venous Thromboembolism

Venous thromboembolism (VTE) has been observed in patients taking tofacitinib in studies and post-marketing reporting. In a study, patients were treated with tofacitinib 5 mg twice daily, tofacitinib 10 mg twice daily or a TNF inhibitor. A dose dependent increase in pulmonary embolism (PE) events was observed in patients treated with tofacitinib compared to TNF inhibitors. Many of these PE events were serious and some resulted in death. PE events were reported more frequently in this study in patients taking tofacitinib relative to other studies across the tofacitinib program.

Deep vein thrombosis (DVT) events were observed in all three treatment groups in the study.

Assess patients for VTE risk factors before starting treatment and periodically during treatment. Use TOZIB with caution in patients in whom risk factors are identified. Urgently evaluate patients with signs and symptoms of VTE. Discontinue tofacitinib while evaluating suspected VTE, regardless of dose or indication.

Mortality

A higher rate of mortality, including sudden cardiovascular (CV) death, is reported in rheumatoid arthritis patients 50 years of age and older, with at least one CV risk factor treated with tofacitinib 10 mg twice a day, compared to those treated with tofacitinib 5 mg given twice daily or TNF blockers.

A dosage of TOZIB 10 mg twice daily is not recommended for the treatment of RA or PsA.

For the treatment of UC, use TOZIB at the lowest effective dose and for the shortest duration needed to achieve/maintain therapeutic response.

Malignancy and Lymphoproliferative Disorder (Excluding Non-melanoma Skin Cancer [NMSC])

Consider the risks and benefits of TOZIB treatment prior to initiating therapy in patients with current or a history of malignancy other than a successfully treated non-melanoma skin cancer (NMSC) or when considering continuing TOZIB in patients who develop a malignancy. The possibility exists for TOZIB to affect host defenses against malignancies.

Lymphomas have been observed in patients treated with tofacitinib. Patients with rheumatoid arthritis, particularly those with highly active disease may be at a higher risk (up to several-fold) than the general population for the development of lymphoma. The role of tofacitinib, in the development of lymphoma is uncertain.

Other malignancies were observed in studies and the post-marketing setting, including, but not limited to, lung cancer, breast cancer, melanoma, prostate cancer and pancreatic cancer.

The role of treatment with tofacitinib on the development and course of malignancies is not known.

Recommendations for non-melanoma skin cancer are presented below.

Rheumatoid Arthritis

Cases of malignancies (excluding NMSC) and lymphoma have been reported in several patients receiving tofacitinib or tofacitinib plus DMARD.

Psoriatic Arthritis

Cases of malignancies (excluding NMSC) has been reported in psoriatic arthritis patients receiving tofacitinib.

Ulcerative Colitis

Cases of malignancies (excluding NMSC) has been reported in psoriatic arthritis patients receiving tofacitinib.

Non-melanoma Skin Cancer

Non-melanoma skin cancers (NMSCs) have been reported in patients treated with tofacitinib. Periodic skin examination is recommended for patients who are at increased risk for skin cancer.

Thrombosis

Thrombosis, including pulmonary embolism, deep venous thrombosis, and arterial thrombosis, have occurred in patients treated with tofacitinib and other Janus kinase (JAK) inhibitors used to treat inflammatory conditions. Patients with rheumatoid arthritis 50 years of age and older with at least one CV risk factor treated with tofacitinib 10 mg twice daily compared to tofacitinib 5 mg twice daily or TNF blockers had an observed increase in incidence of these events. Many of these events were serious and some resulted in death.

A dosage of tofacitinib 10 mg twice daily is not recommended for the treatment of RA.

Cases of pulmonary embolism were reported in patients taking tofacitinib 10 mg twice daily, including one death in a patient with advanced cancer.

Promptly evaluate patients with symptoms of thrombosis and discontinue TOZIB in patients with symptoms of thrombosis.

Avoid TOZIB in patients that may be at increased risk of thrombosis.

Gastrointestinal Perforations

Events of gastrointestinal perforation have been reported in rheumatoid arthritis patients receiving concomitant therapy with non-steroidal anti-inflammatory drugs (NSAIDs) and/or corticosteroids. The role of JAK inhibition in these events is not known. Events were primarily reported as diverticular perforation, peritonitis, abdominal abscess and appendicitis. The relative contribution of these concomitant medications vs. tofacitinib to the development of gastrointestinal perforations is not known.

TOZIB should be used with caution in patients who may be at increased risk for gastrointestinal perforation (e.g., patients with a history of diverticulitis). Patients presenting with new onset abdominal symptoms should be evaluated promptly for early identification of gastrointestinal perforation.

There was no discernable difference in frequency of gastrointestinal perforation between the placebo and the TOZIB arms in clinical trials of patients with UC, and many of them were receiving background corticosteroids.

Hypersensitivity

Reactions such as angioedema and urticaria that may reflect drug hypersensitivity have been observed in patients receiving tofacitinib. Some events were serious. Many of these events occurred in patients that have a history of multiple allergies. If a serious hypersensitivity reaction occurs, promptly discontinue TOZIB while evaluating the potential cause or causes of the reaction.

Laboratory Parameters

Lymphocytes: Lymphocyte counts <500 cells/mm^3 were associated with an increased incidence of treated and serious infections. It is not recommended to initiate TOZIB treatment in patients with a low lymphocyte count (i.e., <500 cells/mm^3). In patients who develop a confirmed absolute lymphocyte count <500 cells/mm^3 treatment with TOZIB is not recommended. Lymphocytes should be monitored at baseline and every 3 months thereafter. For recommended modifications based on lymphocyte counts.

Neutrophils: Treatment with tofacitinib was associated with an increased incidence of neutropenia (<2000 cells/mm^3) compared to placebo. It is not recommended to initiate TOZIB treatment in patients with a low neutrophil count (i.e., ANC <1000 cells/mm^3). For patients who develop a persistent ANC of 500-1000 cells/mm^3, reduce TOZIB dose or interrupt TOZIB dosing until ANC is >1000 cells/mm^3. In patients who develop a confirmed absolute neutrophil count <500 cells/mm^3, treatment with TOZIB is not recommended. Neutrophils should be monitored at baseline and after 4 to 8 weeks of treatment and every 3 months thereafter.

Hemoglobin: It is not recommended to initiate TOZIB treatment in patients with low hemoglobin values (i.e., <9 g/dL). Treatment with TOZIB should be interrupted in patients who develop hemoglobin levels <8 g/dL or whose hemoglobin level drops >2 g/dL on treatment. Hemoglobin should be monitored at baseline and after 4 - 8 weeks of treatment and every 3 months thereafter.

Lipids: Treatment with tofacitinib was associated with increases in lipid parameters such as total cholesterol, low-density lipoprotein (LDL) cholesterol, and high-density lipoprotein (HDL) cholesterol. Maximum effects were generally observed within 6 weeks. Assessment of lipid parameters should be performed approximately 4 - 8 weeks following initiation of TOZIB therapy. Patients should be managed according to current local clinical guidelines for the management of hyperlipidemia. Increase in total and LDL cholesterol associated with TOZIB may be decreased to pre-treatment levels with statin therapy.

Vaccinations

No data are available on the secondary transmission of infection by live vaccines to patients receiving tofacitinib. It is recommended that live vaccines not be given concurrently with TOZIB. It is recommended that all patients be brought up to date with all immunizations in agreement with current immunization guidelines prior to initiating TOZIB therapy. The interval between live vaccinations and initiation of TOZIB therapy should be in accordance with current vaccination guidelines regarding immunomodulatory agents. Consistent with these guidelines, if live zoster vaccine is administered, it should only be administered to patients with a known history of chickenpox or those that are seropositive for varicella zoster virus. Vaccination should occur at least 2 weeks but preferably 4 weeks before initiating immunomodulatory agents such as TOZIB.

Intolerance to excipients

This medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine. This medicinal product contains less than 1 mmol sodium (23 mg) per tablet, that is to say essentially ‘sodium-free’

4.5 Interaction with Other Medicinal Products and Other Forms of Interaction

Table 3 includes drugs with clinically important drug interactions when administered concomitantly with TOZIB and instructions for preventing or managing them.

Table 3: Clinical Relevant Interactions Affecting TOZIB When Co-administered with Other Drugs

Strong CYP3A4 Inhibitors (e.g., ketoconazole)	
Clinical Impact	Increased exposure to tofacitinib
Intervention	Dosage adjustment of TOZIB is recommended
Moderate CYP3A4 Inhibitors Co-administered with Strong CYP2C19 Inhibitors (e.g., fluconazole)	
Clinical Impact	Increased exposure to tofacitinib
Intervention	Dosage adjustment of TOZIB is recommended
Strong CYP3A4 Inducers (e.g., rifampin)	
Clinical Impact	Decreased exposure to tofacitinib and may result in loss of or reduced clinical response
Intervention	Co-administration with TOZIB is not recommended
Immunosuppressive Drugs (e.g., azathioprine, tacrolimus, cyclosporine)	
Clinical Impact	Risk of added immunosuppression; co-administration with biologic DMARDs or potent immunosuppressants has not been studied in patients with rheumatoid arthritis
Intervention	Co-administration with TOZIB is not recommended

4.6 Fertility, pregnancy and lactation

Pregnancy

Risk Summary

Available data with tofacitinib use in pregnant women are insufficient to establish a drug associated risk of major birth defects, miscarriage or adverse maternal or fetal outcomes. There are risks to the mother and the fetus associated with rheumatoid arthritis in pregnancy. In animal reproduction studies, feticidal and teratogenic effects were noted when pregnant rats and rabbits received tofacitinib during the period of organogenesis at exposures multiples of 73 times. Further, in a peri and post-natal study in rats, tofacitinib resulted in reductions in live litter size, postnatal survival, and pup body weights at exposure multiples of approximately 73 times the recommended dose of 5 mg twice daily.

Women of reproductive potential should be advised to use effective contraception during treatment with tofacitinib and for at least 4 weeks after the last dose.

The estimated background risks of major birth defects and miscarriage for the indicated populations are unknown. All pregnancies have a background risk of birth defect, loss, or other adverse outcomes.

Clinical Considerations

Disease-associated Maternal and/or Embryo/Fetal Risk
Published data suggest that increased disease activity is associated with the risk of developing adverse pregnancy outcomes in women with rheumatoid arthritis. Adverse pregnancy outcomes include preterm delivery (before 37 weeks of gestation), low birth weight (less than 2500 g) infants, and small for gestational age at birth.

Lactation

Risk Summary

There are no data on the presence of tofacitinib in human milk, the effects on a breastfed infant, or the effects on milk production. Tofacitinib is present in the milk of lactating rats. When a drug is present in animal milk, it is likely that the drug will be present in human milk. Given the serious adverse reactions seen in patients treated with tofacitinib, such as increased risk of serious infections, advise patients that breastfeeding is not recommended during treatment and for at least 18 hours after the last dose of TOZIB.

Females and Males of Reproductive Potential

Contraception

Females
In an animal reproduction study, tofacitinib at AUC multiples of 13 times the recommended dose of 5 mg twice daily demonstrated adverse embryo-fetal findings.

However, there is uncertainty as to how these animal findings relate to females of reproductive potential treated with the recommended clinical dose. Consider pregnancy planning and prevention for females of reproductive potential.

Infertility

Females
Based on findings in rats, treatment with tofacitinib may result in reduced fertility in females of reproductive potential. It is not known if this effect is reversible.

4.7 Effects on Ability to Drive and Use Machines

No formal studies have been conducted on the effects on the ability to drive and use machines.

4.8 Undesirable Effects

Summary of the safety profile

Rheumatoid arthritis

The most common serious adverse reactions were serious infections. The most common serious infections reported with tofacitinib were pneumonia, herpes zoster, urinary tract infection, cellulitis, diverticulitis, and appendicitis. Among opportunistic infections, TB and other mycobacterial infections, cryptococcus, histoplasmosis, esophageal candidiasis, multidermatomal herpes zoster, cytomegalovirus infection, BK virus infections and listeriosis were reported with tofacitinib. Some patients have presented with disseminated rather than localized disease. Other serious infections may also occur (e.g., coccidioidomycosis).

The most commonly reported adverse reactions during study were headache, upper respiratory tract infections, viral upper respiratory tract infection, diarrhea, nausea and hypertension.

Psoriatic arthritis

Overall, the safety profile observed in patients with active PsA treated with tofacitinib was consistent with the safety profile observed in patients with RA treated with tofacitinib.

Ankylosing spondylitis

Overall, the safety profile observed in patients with active AS treated with tofacitinib was consistent with the safety profile observed in patients with RA treated with tofacitinib.

Ulcerative colitis

The most commonly reported adverse reactions in patients receiving tofacitinib 10 mg twice daily were headache, nasopharyngitis, nausea, and arthralgia.

The most common categories of serious adverse reactions were gastrointestinal disorders and infections, and the most common serious adverse reaction was worsening of UC.

Overall, the safety profile observed in patients with UC treated with tofacitinib was consistent with the safety profile of tofacitinib in the RA indication.

Tabulated list of adverse reactions

The adverse reactions listed in the table below are from clinical studies in patients with RA, PsA, AS and UC and are presented by System Organ Class (SOC) and frequency categories, defined using the following convention: very common, common, uncommon, rare, very rare, or not known. Within each frequency grouping, adverse reactions are presented in the order of decreasing seriousness.

Table 4: Adverse reactions

System organ class	Common	Uncommon	Rare	Very rare	Not known
Infections and infestations	Pneumonia Influenza Herpes zoster Urinary tract infection Sinusitis Bronchitis Nasopharyngitis Pharyngitis	Tuberculosis Diverticulitis Pyelonephritis Cellulitis Herpes simplex Gastroenteritis viral Viral infection	Sepsis Urosepsis Disseminated TB Bacteremia *Pneumocystis jirovecii* pneumonia Pneumonia pneumococcal Pneumonia bacterial Cytomegalovirus infection Arthritis bacterial	Tuberculosis of central nervous system Meningitis cryptococcal Necrotizing fasciitis Encephalitis Staphylococcal bacteremia *Mycobacterium avium* complex infection Atypical mycobacterial infection	
Neoplasms benign, malignant and unspecified (incl cysts and polyps)		Lung cancer Non-melanoma skin cancers	Lymphoma		
Blood and lymphatic system disorders	Lymphopenia Anaemia	Leukopenia Neutropenia			
Immune system disorders					Hypersensitivity Angioedema Urticaria

Metabolism and nutrition disorders		Dyslipidemia Hyperlipidaemia Dehydration			
Psychiatric disorders		Insomnia			
Nervous system disorders	Headache	Paraesthesia			
Vascular disorders	Hypertension	Venous thromboembolism **			
Cardiac disorders		Myocardial infarction			
Respiratory, thoracic and mediastinal disorders	Cough	Dyspnoea Sinus congestion			
Gastrointestinal disorders	Abdominal pain Vomiting Diarrhea Nausea Gastritis Dyspepsia				
Hepatobiliary disorders		Hepatic steatosis Hepatic enzyme increased Transaminases increased Gamma glutamyl-transferase increased	Liver function test abnormal		
Skin and subcutaneous tissue disorders	Rash	Erythema Pruritus			
Musculoskeletal and connective tissue disorders	Arthralgia	Joint swelling Tendonitis	Musculoskeletal pain		
General disorders and administration site conditions	Oedema peripheral	Pyrexia Fatigue			
Investigations	Blood creatine phosphokinase increased	Blood creatinine increased Blood cholesterol increased Low density lipoprotein increased Weight increased			
Injury, poisoning and procedural complications		Ligament sprain Muscle strain			

**Venous thromboembolism includes PE, DVT, and Retinal Venous Thrombosis

Description of selected adverse reactions

Venous thromboembolism (VTE)

Rheumatoid arthritis

VTE has been observed at an increased rate in patients who were 50 years of age and older and had at least one additional CV risk factor and treated with tofacitinib as compared to TNF inhibitors. The majority of these events were serious and some resulted in death.

Ankylosing spondylitis

No VTE events were reported in patients receiving tofacitinib up to 48 week.

Ulcerative colitis (UC)

Cases of PE and DVT have been observed in patients using tofacitinib 10 mg twice daily and with underlying VTE risk factor(s).

Overall infections

Rheumatoid arthritis

Cases of infections over 0-3 months were reported in the 5 mg twice daily and 10 mg twice daily tofacitinib monotherapy. An increased risk of the rates of infections over 0-3 months is seen with concomitant use of 5 mg twice daily and 10 mg twice daily tofacitinib plus DMARD.

The most commonly reported infections were upper respiratory tract infections and nasopharyngitis.

Ankylosing spondylitis

In the clinical studies, during the placebo-controlled period of up to 16 weeks, the frequency of infections in the tofacitinib 5 mg twice daily group was 27.6% and the frequency in the placebo group was 23.0%.

Ulcerative colitis

In the entire treatment experience with tofacitinib, the most commonly reported infection was nasopharyngitis.

Serious infections

Rheumatoid arthritis

The most common serious infections included pneumonia, herpes zoster, urinary tract infection, cellulitis, gastroenteritis and diverticulitis. Cases of opportunistic infections have been reported

Ankylosing spondylitis (AS)

In the studies, there was one serious infection (aseptic meningitis) reported.

Ulcerative colitis (UC)

The incidence rates and types of serious infections in the UC clinical studies were generally similar to those reported in RA clinical studies with tofacitinib monotherapy treatment groups.

Serious infections in the elderly

The frequency of serious infection among tofacitinib-treated patients 65 years of age and older was higher than those under the age of 65. As there is a higher incidence of infections in the elderly population in general, caution should be used when treating the elderly.

Serious infections from non-interventional post approval safety study

Data from a non-interventional post approval safety study that evaluated tofacitinib in RA patients from a registry showed that a numerically higher incidence rate of serious infection was observed for the 11 mg prolonged-release tablet administered once daily than the 5 mg film-coated tablet administered twice daily. Data is based on a small number of patients with events observed with relatively large confidence intervals and limited follow up time.

Viral reactivation

Patients treated with tofacitinib who are Japanese or Korean, or patients with long standing RA who have previously received two or more biological DMARDs, or patients with an ALC less than 1,000 cells/mm^3, or patients treated with 10 mg twice daily may have an increased risk of herpes zoster.

In a safety study in patients with RA who were 50 years or older with at least one additional cardiovascular risk factor, an increase in herpes zoster events was observed in patients treated with tofacitinib compared to TNF inhibitors.

Laboratory tests

Lymphocytes

Confirmed ALC less than 750 cells/mm^3 were associated with an increased incidence of serious infections.

In the clinical studies in UC, changes in ALC observed with tofacitinib treatment were similar to the changes observed in clinical studies in RA.

Neutrophils

In the clinical studies in UC, changes in ANC observed with tofacitinib treatment were similar to the changes observed in clinical studies in RA.

Platelets

Patients in the Phase 3 controlled clinical studies (RA, PsA, AS, UC) were required to have a platelet count $\geq$ 100,000 cells/mm^3 to be eligible for enrolment, therefore, there is no information available for patients with a platelet count < 100,000 cells/mm3 before starting treatment with tofacitinib.

Liver enzyme tests

Confirmed increases in liver enzymes greater than 3 times the upper limit of normal (3x ULN) were uncommonly observed in RA patients. In those patients experiencing liver enzyme elevation, modification of treatment regimen, such as reduction in the dose of concomitant DMARD, interruption of tofacitinib, or reduction in tofacitinib dose, resulted in decrease or normalisation of liver enzymes.

In the clinical studies in UC, changes in liver enzyme tests observed with tofacitinib treatment were similar to the changes observed in clinical studies in RA.

Lipids

Elevations in lipid parameters (total cholesterol, LDL cholesterol, HDL cholesterol, triglycerides) were observed one month after the initiation of tofacitinib therapy and remained stable thereafter.

Upon withdrawal of tofacitinib treatment, lipid levels returned to baseline.

Mean LDL cholesterol/HDL cholesterol ratios and Apolipoprotein B (ApoB)/ApoA1 ratios were essentially unchanged in tofacitinib-treated patients.

In an RA controlled clinical study, elevations in LDL cholesterol and ApoB decreased to pretreatment levels in response to statin therapy.

In the RA long-term safety populations, elevations in the lipid parameters remained consistent with what was seen in the controlled clinical studies.

In the clinical studies in UC, changes in lipids observed with tofacitinib treatment were similar to the changes observed in clinical studies in RA.

Myocardial infarction

Rheumatoid arthritis

Few fatal myocardial infarctions were reported with rates similar in patients treated with tofacitinib compared to TNF inhibitors.

Malignancies excluding NMSC

Rheumatoid arthritis

The incidence rates (95% CI) for lymphoma for tofacitinib 5 mg twice daily, tofacitinib 10 mg twice daily, and TNF inhibitors were 0.07, 0.11, and 0.02 patients with events per 100 patient-years, respectively.

Paediatric population

Polyarticular juvenile idiopathic arthritis and juvenile PsA

The adverse reactions in JIA patients in the clinical development program were consistent in type and frequency with those seen in adult RA patients, with the exception of some infections (influenza, pharyngitis, sinusitis, viral infection) and gastrointestinal or general disorders (abdominal pain, nausea, vomiting, pyrexia, headache, cough), which were more common in JIA paediatric population. MTX was the most frequent concomitant csDMARD used. There are insufficient data regarding the safety profile of tofacitinib used concomitantly with any other csDMARDs.

Infections

In the study, infection was the most commonly reported adverse reaction. The infections were generally mild to moderate in severity.

In the integrated safety population, serious infections during treatment with tofacitinib have been reported: pneumonia, epidural empyema (with sinusitis and subperiosteal abscess), pilonidal cyst, appendicitis, escherichia pyelonephritis, abscess limb, and UTI.

Hepatic events

Patients in the JIA pivotal study were required to have AST and ALT levels less than 1.5 times the upper limit of normal to be eligible for enrolment. In the integrated safety population, there were 2 patients with ALT elevations ≥ 3 times the ULN at 2 consecutive visits. Neither event met Hy's Law criteria. Both patients were on background MTX therapy and each event resolved after discontinuation of MTX and permanent discontinuation of tofacitinib.

Laboratory tests

Changes in laboratory tests in JIA patients in the clinical development program were consistent with those seen in adult RA patients.

4.9 Overdose

There is no specific antidote for overdose with tofacitinib. In case of an overdose, it is recommended that the patient be monitored for signs and symptoms of adverse reactions.

In a study in subjects with end stage renal disease (ESRD) undergoing hemodialysis, plasma tofacitinib concentrations declined more rapidly during the period of hemodialysis and dialyzer efficiency. However, due to the significant non-renal clearance of tofacitinib, the fraction of total elimination occurring by hemodialysis was small, and thus limits the value of hemodialysis for treatment of overdose with tofacitinib.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Mechanism of Action

Tofacitinib is a Janus kinase (JAK) inhibitor. JAKs are intracellular enzymes which transmit signals arising from cytokine or growth factor-receptor interactions on the cellular membrane to influence cellular processes of hematopoiesis and immune cell function. Within the signaling pathway, JAKs phosphorylate and activate Signal Transducers and Activators of Transcription (STATs) which modulate intracellular activity including gene expression. Tofacitinib modulates the signaling pathway at the point of JAKs, preventing the phosphorylation and activation of STATs. JAK enzymes transmit cytokine signaling through pairing of JAKs (e.g., JAK1/JAK3, JAK1/JAK2, JAK1/TyK2, JAK2/JAK2). Tofacitinib inhibited the in vitro activities of JAK1/JAK2, JAK1/JAK3, and JAK2/JAK2 combinations with IC_{50} of 406, 56, and 1377 nM, respectively. However, the relevance of specific JAK combinations to therapeutic effectiveness is not known.

Pharmacodynamics

Treatment with tofacitinib was associated with dose-dependent reductions of circulating CD16/56+ natural killer cells, with estimated maximum reductions occurring at approximately 8-10 weeks after initiation of therapy. These changes generally resolved within 26 weeks after discontinuation of treatment. Treatment with tofacitinib was associated with dose-dependent increases in B cell counts. Changes in circulating T-lymphocyte counts and T-lymphocyte subsets (CD3+, CD4+ and CD8+) were small and inconsistent. The clinical significance of these changes is unknown.

Total serum IgG, IgM, and IgA levels after 6-month dosing in patients with rheumatoid arthritis were lower than placebo; however, changes were small and not dose-dependent.

After treatment with tofacitinib in patients with rheumatoid arthritis, rapid decreases in serum C-reactive protein (CRP) were observed and maintained throughout dosing. Changes in CRP observed with tofacitinib treatment do not reverse fully within 2 weeks after discontinuation, indicating a longer duration of pharmacodynamic activity compared to the pharmacokinetic half-life.

Similar changes in T cells, B cells, and serum CRP have been observed in patients with active psoriatic arthritis although reversibility was not assessed. Total serum immunoglobulins were not asserted in patients with active psoriatic arthritis.

5.2 Pharmacokinetic Properties

Following oral administration of tofacitinib, peak plasma concentration is reached within 0.5-1 hour, elimination half-life is about 3 hours and a dose-proportional increase in systemic exposure was observed in the therapeutic dose range. Steady state concentration is achieved in 24-48 hours with negligible accumulation after twice daily administration.

Absorption

The absolute oral bioavailability of tofacitinib is 74%. Co-administration of tofacitinib with a high-fat meal resulted in no changes in AUC while C_{max} was reduced by 32%. In clinical trials, tofacitinib was administered without regard to meals.

Distribution

After intravenous administration, the volume of distribution is 87 L. The protein binding of tofacitinib is approximately 40%. Tofacitinib binds predominantly to albumin and does not appear to bind to α1-acid glycoprotein. Tofacitinib distributes equally between red blood cells and plasma.

Metabolism and Excretion

Clearance mechanisms for tofacitinib are approximately 70% hepatic metabolism and 30% renal excretion of the parent drug. The metabolism of tofacitinib is primarily mediated by CYP3A4 with minor contribution from CYP2C19. In a human radiolabeled study, more than 65% of the total circulating radioactivity was accounted for by unchanged tofacitinib, with the remaining 35% attributed to 8 metabolites, each accounting for less than 8% of total radioactivity. The pharmacologic activity of tofacitinib is attributed to the parent molecule.

6 PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

<u>Tablet core:</u>

Microcrystalline Cellulose, Lactose Monohydrate, Croscarmellose Sodium, Magnesium Stearate

<u>Film coat:</u>

Opadry II Complete Film Coating System 33G28523 White

6.2 Incompatibilities

Not applicable

6.3 Shelf Life

Please refer to the expiry date on the product labels.

6.4 Special Precautions for Storage

Store below 30°C.

6.5 Nature and Content of Container

Alu/Alu blister of 1x10's, 3x10's, 6x10's, 10x10's, 50x10's, 100x10's, 1x14's, 2x14's, 4x14's, 6x14's, 8x14's, 12x14's and 100x14's tablets packed in the outer carton. Not all pack size will be marketed.

6.6 Special precautions for disposal and other handling

No special requirements.

7 MANUFACTURER

Manufactured by:
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8 DATE OF REVISION

13/03/2024

Package insert is drafted in Times New Roman with font size of 11, subject to change based on any other legible font and font size.